



Structured & Cybernetic Models of Microbial Growth

Going beyond Monod: giving the cell an internal state and a strategy

Biochemical Engineering · Based on Villadsen, Nielsen & Lidén, Bioreaction Engineering Principles (3rd ed.), Sect. 7.4, and the cybernetic-modelling literature of Ramkrishna and co-workers

Why unstructured models are not enough

Monod-type kinetics describe balanced growth only — the cell is a single variable, biomass X

Unstructured (Ch. 7.3)

- Biomass is one lumped quantity; $\mu = f(s)$ responds instantly to the environment
- Implies a fixed cell composition, so it can only be valid for balanced growth
- Cannot describe: lag phases, shift-up / shift-down transients, diauxic growth, changing RNA/protein content with μ

Structured (Ch. 7.4)

- Biomass split into ≥ 2 components (compartments, key enzymes, cybernetic variables)
- Intracellular state carries memory of the past $\rightarrow$ transients emerge naturally
- 'Simple' structure: a few well-chosen variables, not a full metabolic network

Model classification (after Tsuchiya, Fredrickson & Aris)

Unstructured, non-segregated

Monod, Contois — the workhorse

Structured, non-segregated

Compartment & cybernetic models
 $\leftarrow$ this lecture

Unstructured, segregated

Cell-size / age distributions

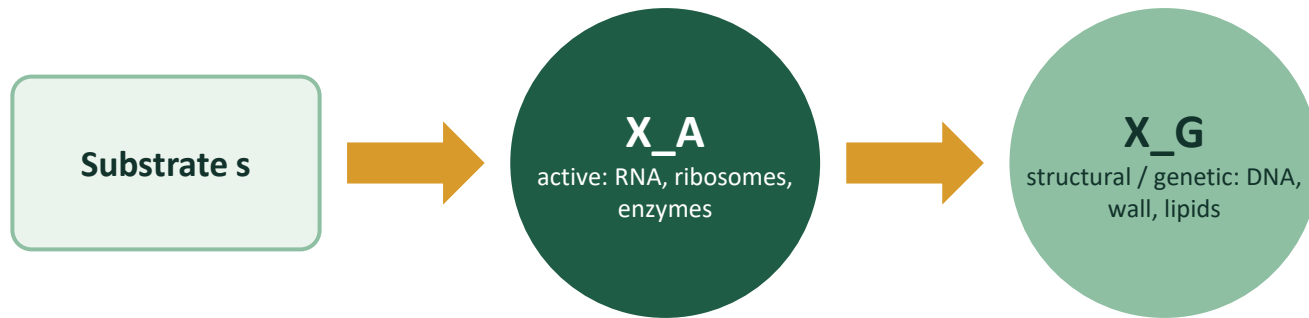
Structured, segregated

Population balances with internal state

Rows: cell composition lumped vs. resolved. Columns: population treated as average cell vs. distribution of individuals.

Compartment models: the two-compartment idea

Williams (1967) — the first structured model; Nielsen et al. (1991) — the version in the textbook



Reaction 1: $s \rightarrow X_A$ (rate r_1 , e.g. Monod in s)
Reaction 2: $X_A \rightarrow X_G$ (rate r_2 , first order in X_A)

Key modelling choices

- Biomass = $X_A + X_G$, with mass fractions $x_A + x_G = 1$
- Only the active compartment X_A synthesises new material — it is the 'growth machinery'
- Rates are written per unit biomass; a dilution term $-\mu \cdot x_A$ appears because the cell volume grows
- Two ODEs for the fractions replace one for X ; parameters stay few (k, K_s, Y)
- Any structured model must reduce to a Monod-like law at balanced growth — a useful consistency check

What structure buys you

Balance for the active compartment and its consequences

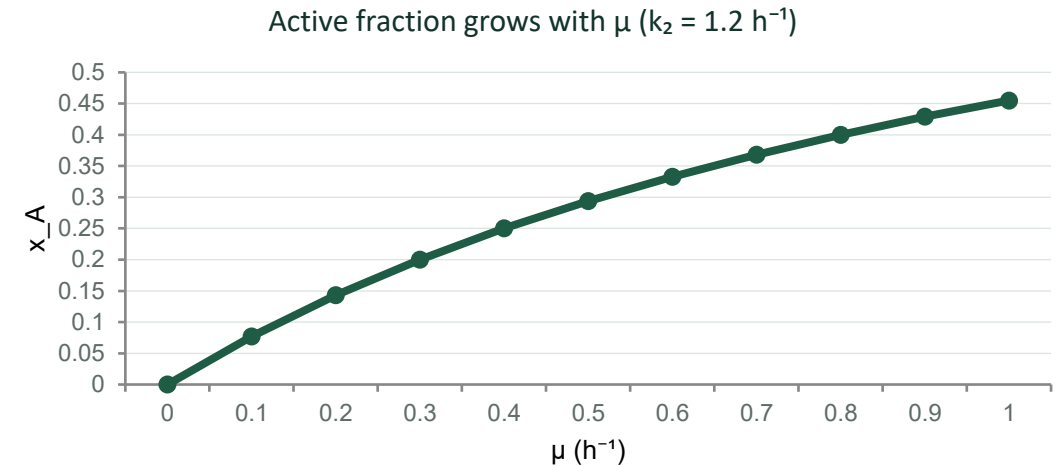
Balance on the active compartment (per unit biomass):

$$dx_A/dt = r_1 - r_2 - \mu \cdot x_A$$

$$\begin{aligned} r_1 &= k_1 \cdot s / (K_1 + s) \cdot x_A && \text{(synthesis)} \\ r_2 &= k_2 \cdot x_A && (\rightarrow \text{structure}) \\ \mu &= r_1 && \text{(net growth)} \end{aligned}$$

Balanced growth, $dx_A/dt = 0$:

$$x_A = \mu / (\mu + k_2) \Rightarrow \mu = \mu_{\max} \cdot s / (K + s)$$



1 Composition vs. μ

RNA / ribosome content rises roughly linearly with growth rate — matches chemostat data for E. coli, yeast, and many others.

2 Lag phase

Cells from a slow-growth state have little X_A ; after a shift-up the machinery must first be built, so growth lags even though s is high.

3 Shift-down / washout

After a drop in s , X_A is 'too large' and is diluted only slowly — the model overshoots and then relaxes, as observed.

Cybernetic models: the cell as an optimal strategist

Ramkrishna & co-workers (Purdue), from Kompala et al. 1984 onwards

The core idea

Instead of modelling the regulatory mechanism (repressors, cAMP, inducers ...), assume the cell allocates scarce resources so as to maximise an objective — typically its instantaneous growth rate. Regulation is then the solution of an optimisation problem, not a set of extra kinetic terms.

Why bother?

- Mechanistic regulation is rarely known well enough to write kinetics for
- Every new inhibition/induction term adds parameters
- Optimality laws add almost no parameters and stay stable across phenotypes



Enzyme synthesis

Fraction of the 'induction budget' spent on enzyme E_i .
Matching law: $u_i = r_i / \sum_j r_j$



Enzyme activity

Degree to which existing E_i is switched on. Proportional law: $v_i = r_i / \max_j r_j$

r_i = rate of growth supported by substrate i alone (single-substrate Monod kinetics $\times$ enzyme level). Only u_i and v_i are new — no new parameters.

Worked case: diauxic growth on two sugars

Kompala-type model, two substrates, two inducible enzymes — simulated

Model equations ($i = 1, 2$)

$$r_i = \mu_{i,\max} \cdot e_i \cdot s_i / (K_i + s_i)$$

$$ds_i/dt = -v_i \cdot r_i \cdot X / Y_i$$

$$de_i/dt = \alpha_i \cdot u_i \cdot s_i / (K_i + s_i) + e^* - (\beta + \mu) \cdot e_i$$

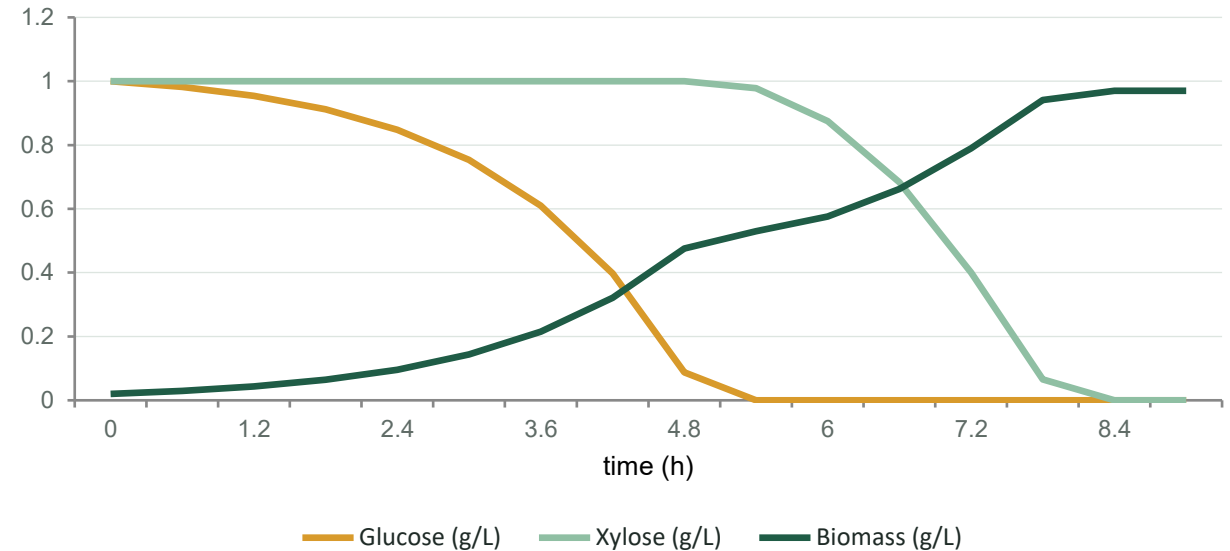
$$dX/dt = \mu X, \quad \mu = \sum_i v_i \cdot r_i$$

$$u_i = r_i / \sum_j r_j \quad (\text{matching})$$

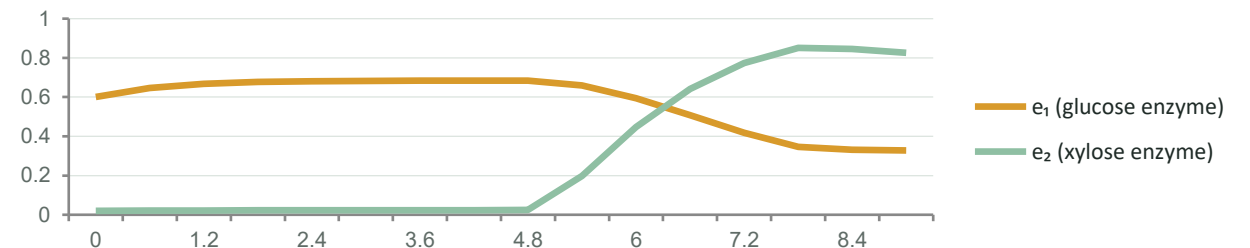
$$v_i = r_i / \max_j r_j \quad (\text{proportional})$$

e_i = relative enzyme level; e^* small constitutive synthesis; β = enzyme decay.
Parameters here: $\mu_{\max} = 1.0 / 0.45 \text{ h}^{-1}$, $K = 0.01 / 0.05 \text{ g L}^{-1}$, $Y = 0.5 / 0.45$.

Sequential utilisation with an intermediate lag



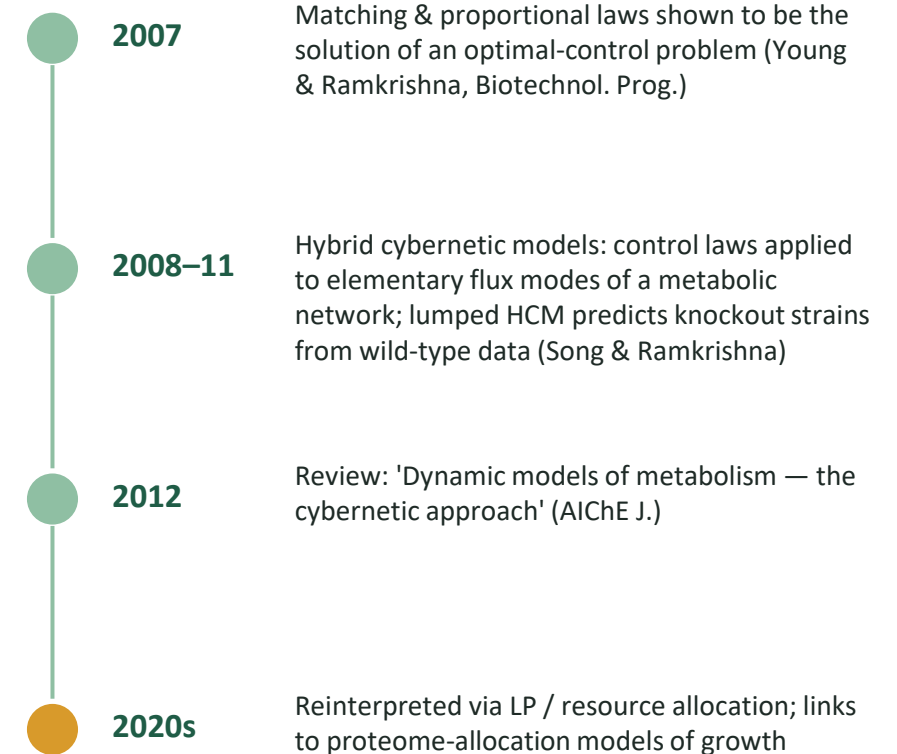
Enzyme levels: e_2 is only built once glucose is gone ($u_2 \approx 0$ before)



Compartment vs. cybernetic, and what came next

	Compartment models	Cybernetic models
What is 'structured'	Biomass split into 2–3 pools (active / structural)	Key enzymes + control variables u_i , v_i
Regulation	Implicit — rates depend on pool sizes	Explicit — optimal allocation of synthesis & activity
Extra parameters	A few rate constants	Enzyme synthesis/decay (α , β); u , v add none
Best at	Lag, shift-up/down, composition vs μ	Mixed substrates, diauxie, switching between pathways
Weakness	Compartments are abstractions, hard to measure	Objective function is a hypothesis; can miss non-optimal behaviour

Cybernetic modelling since the textbook



Take-aways

1 Structure = memory

Giving the cell internal variables lets transients (lag, shift, diauxie) emerge instead of being bolted on.

2 Keep it simple

Two compartments or two control variables already capture most of the qualitative behaviour with few parameters.

3 Optimality as a modelling tool

Cybernetic laws replace unknown regulatory mechanisms by a hypothesis about what the cell is trying to do — testable and cheap.

4 Check the limit

Every structured model should collapse to a Monod-like law under balanced growth.

*Discussion question: Monod's *E. coli* grow on glucose + lactose. Which model — compartment or cybernetic — would you use to predict the length of the diauxic lag, and what would you need to measure to fit it?*

The biomass reaction is a pseudo-reaction

It converts precursor metabolites into one unit of cell; its flux is the predicted growth rate (μ).

Precursor pools

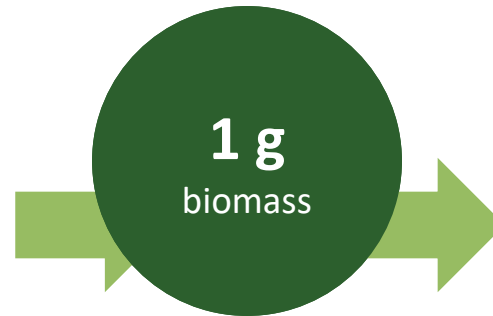
Amino acids

Nucleotides

Fatty acids

Sugars

Inorganic ions



$v_{\text{biomass}} = \mu$

Macromolecules

Protein

RNA / DNA

Lipids

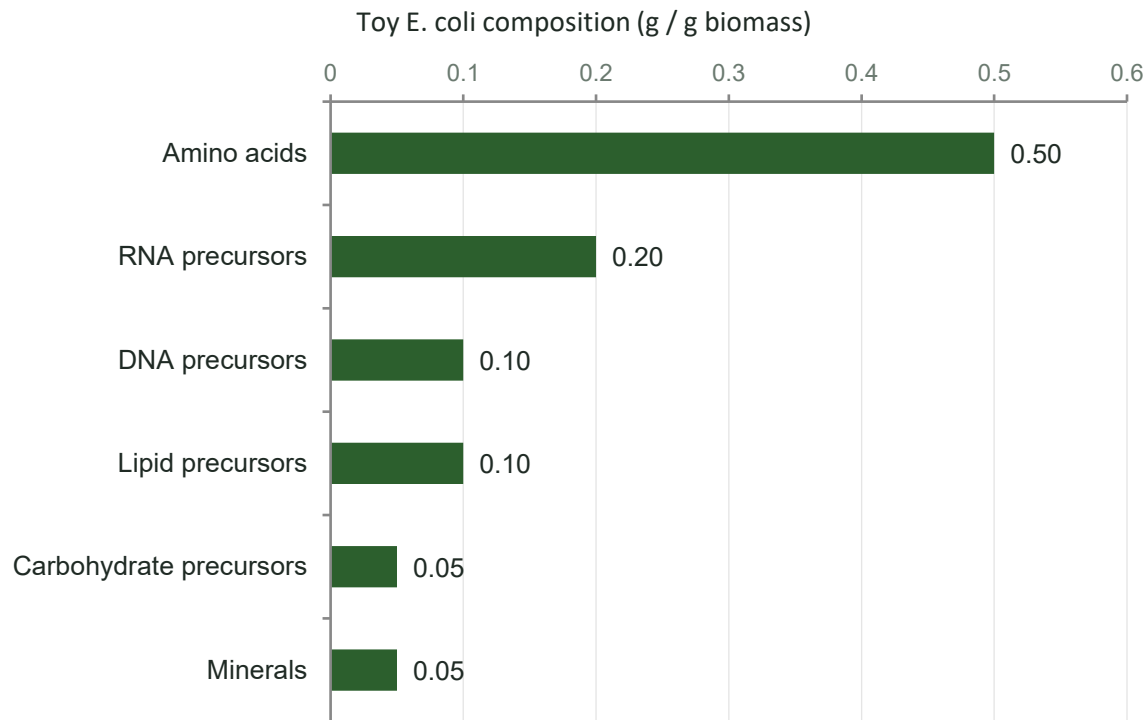
Cell wall

Cofactors

Toy quantitative version: 0.50 g amino acids + 0.20 g RNA precursors + 0.10 g DNA precursors + 0.10 g lipid precursors + 0.05 g carbohydrate precursors + 0.05 g minerals $\rightarrow$ 1.00 g biomass

The coefficients are not arbitrary

Each coefficient is the amount of precursor needed per unit biomass, from measured cell composition.



Consequence for fluxes

If RNA precursors are 0.20 g / g biomass, then whenever $v_{\text{biomass}} > 0$ the model must carry at least $0.20 \times \mu$ through nucleotide synthesis. Growth pulls on every precursor pathway in fixed ratios.

Genome-scale: pools become metabolites

$0.45 \text{ Ala} + 0.30 \text{ Gly} + \dots + 0.12 \text{ ATP} + 0.10 \text{ GTP} + \dots + 0.03 \text{ palmitate} + \dots \rightarrow \text{Biomass}$
Same idea, hundreds of terms (mmol / gDW).

Take-away: the biomass reaction ties growth to a fixed demand on every precursor pathway; its coefficients encode what a cell is made of.